Research Readiness Statement

a. Degree program and mentor preparation. Colton Goodrich is a PhD student in Geosensing Systems Engineering & Sciences at the University of Houston, advised by the proposal PI, Craig L. Glennie. Graduate coursework in remote sensing and geosensing, together with prior degrees in Geology (M.S. and B.S., Brigham Young University) and a Geospatial Analysis certificate (Weber State University), provide a foundational understanding of terrain analysis, digital elevation models, and remote sensing relevant to the proposed research. The proposed project requires the Python scientific stack (NumPy, GDAL, Rasterio, PyTorch), PDAL, WhiteboxTools, and QGIS, together with DEM differencing, co-registration (ICP), and robust statistics; the FI is already proficient in each of these tools and methods. Two areas require further preparation — hierarchical/Bayesian statistical modeling and cold-regions (Antarctic) process geomorphology; the FI will address them through graduate coursework and directed study with the PI before the analyses that depend on them.

b. Graduate study timeline. Degree type: PhD. Subject area: Geosensing Systems Engineering & Sciences. Enrollment: enrolled since August 2025 (approximately one year at the proposal due date). Estimated graduation: August 2029.

c. Other relevant experience. The FI independently designed and built an end-to-end airborne-lidar terrain-analysis pipeline on the Appalachian Plateau — point-cloud processing (PDAL), bare-earth derivative generation (GDAL/WhiteboxTools), hand-labeled training-set construction (QGIS), U-Net and instance-segmentation model training (PyTorch), and DEM-of-Difference change analysis with ICP co-registration and NMAD-based uncertainty — an independent research project that exercises the methods used in the proposed work. Additional relevant experience includes InSAR-based DEM generation and accuracy assessment against lidar references; five years as an Environmental Scientist at the Utah Division of Oil, Gas & Mining (geospatial data analysis and Python automation); prior work as an Earth Scientist at Chevron; a graduate teaching assistantship at Brigham Young University; and a Python Programming course (The Tech Academy, 2024).